NCOS Bird Survey Data Management, QA/QC and eBird Upload Protocol

1. Download raw data file (Feature Class) with the file name format "NCOS_Bird_Observations_YYYYMMDD" from ArcGIS Online into the NCOS_Bird_Observations_Data geodatabase on the CCBER data store. This is to backup the raw data onto the data store and can also be done using ArcPro or ArcMap.

2. In ArcGIS Online, export the raw Bird Observations feature layer in Excel format, then download to appropriate data store or local directory. Add the word "raw" to the end of the file name.

3. Open the "raw" file in Excel (or google sheets) and prepare data for summary:
   3.1. Sort the raw data by the Species column in ascending order to show any Blank species observations. There should be 2 to 4 Blank observation rows that contain only the start and end weather and other data about the survey, including observers. However, this information may also be in the first Species observation of the survey for either of the two survey teams. Create a new sheet tab in the Excel file and name it "Survey_Metadata". In the raw data sheet, copy the 2 to 4 rows containing survey weather data, etc. to the Survey_Metadata sheet. In the Survey_Metadata sheet, remove all columns related to species observation info.

   3.2. In the raw data sheet, sort by the OBJECTID column (***this is not needed if the Creator column is working properly***). In the Observers column, copy the observer names (or initials) from the first entry (these should be in the first row of data, i.e. row 2) and paste into all empty cells in that column up to where the observer names of the other team are entered. Then copy the names of the other team in all empty cells beneath that entry.

   3.3. Filter the data to show only the Blank and NOT LISTED entries for the Species column. These should have some information in the Observation Notes column such as a species common name. Move the common name in the Observation Notes column into the Species column for the Blank or NOT LISTED observations, and correct capitalization, dashes, etc. of the name as needed. ***Also make sure the names are consistent!*** You may remove the common names from the Observation Notes column after moving them to the Species column.
      3.3.1. The species names that were not in the pre-set list and therefore resulted in Blank or NOT LISTED entries should be added to the "Species" domain for the Bird_Observations_Template Feature Class (this is best/easiest to do in ArcPro). This will add them to the pre-set list for the subsequent surveys.

   3.4. Check the Observation Notes column for any typos or incorrect entries and remove them.

4. Create a pivot table of all of the raw data in a separate sheet tab:

4.1. Choose “Species” for the Rows (should automatically sort in ascending order, but check that it is).

4.2. Choose “Repeat Observation” AND “Observers” or “Creator” for the Columns.

4.3. Choose “Count” for the Values and make sure it is the SUM value.

4.4. Choose “Species” for the Filter and make sure that “Blank” is unchecked (or you can delete the 2 to 4 rows copied to the Survey_Metadata sheet from the raw data sheet).

4.5. The above will result in a pivot table with 8 columns, starting with the list of species in the first column on the left. The first 3 count columns summarize the original observations (i.e. “No” for Repeat Observation) for each team and the subtotal, while the second set of 3 columns summarize the the Repeat Observations, and the final column is a grand total. The bottom row is also a grand total.

5. Check Repeat Observations and/or other questionable observations:

5.1. You may need to verify Repeat Observations, as well as other questionable observations, by checking the location of these observations on the map. This is easiest to do on ArcGIS Online. It may be that a Repeat Observation is in fact not, and should therefore be changed from “Yes” to “No” in the raw data sheet. This will automatically change the data in the Pivot Table (though in Excel you may need to refresh the table).

6. Create total count list from the pivot table:

6.1. After verifying and correcting any repeat and other observations, copy the species column and the column with the total count of non-repeat observations for each species to a separate, blank sheet tab and name it “Total_Count”.

7. Prepare eBird template CSV and upload:

7.1. Create a copy of the eBird CSV template (located here on the CCBER data store: \\smb.ccber.ucsb.edu\ccber-data\Ecosystem_Management\Management_Areas\NCOS\Monitoring\Birds\Monthly_Surveys\ebird), and name it “NCOS_Bird_Survey_YYYYMMDD_ebird”.

7.2. Copy the species names (excluding the header row) in the “Total_Count” sheet to the “Common Name” column (column A) in the eBird CSV template. Then copy the “count” column data in the “Total_Count” sheet to the “Number” column (column D) in the eBird CSV template. The “Genus”, “Species”, and “Species Comments” columns will be left blank.

7.3. Enter Survey Metadata information in the first row for the “Date” (column I), “Start Time” (column J), “Number of Observers” (column N), “Duration” (in minutes - column O), and “Submission Comments” (column S). The info for all of the other metadata columns (“Location Name”, “Latitude”, “Longitude”, “State/Province”, "Country Code", "Protocol", "All Observations Reported", "Effort Distance Miles", and "Effort area acres") should be prefilled. Once you've completed entering the survey metadata in the first row, copy this row (only the metadata columns) into all rows of the species list.

7.4. After double-checking that all information is correct, remove the first column (with the column headers), and save as a CSV file (if not already a CSV, otherwise just save it). Make sure that a copy of this file is saved in this data store location: \\smb.ccber.ucsb.edu\ccber-data\Ecosystem_Management\Management_Areas\NCOS\Monitoring\Birds\Monthly_Surveys\ebird

7.5. Upload to eBird. This process is similar to what is outlined in the "ebird_data_import_process" pdf in the same data store location as the ebird CSV template.